

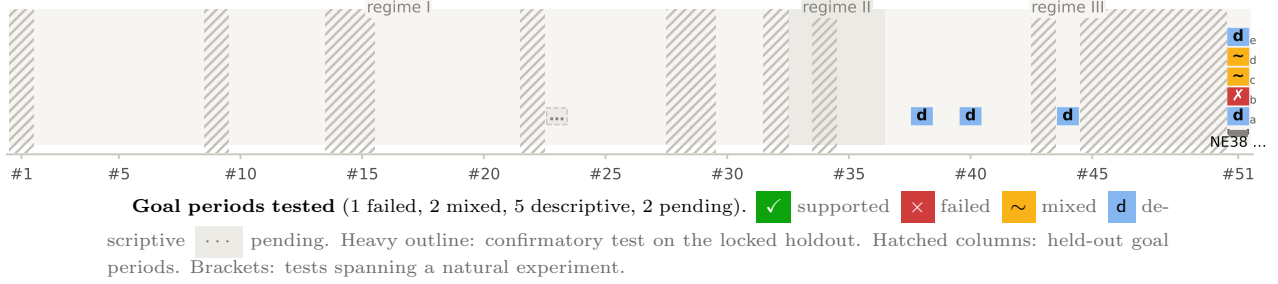
## H22: Private, conflicting goals make #51 a spin glass

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

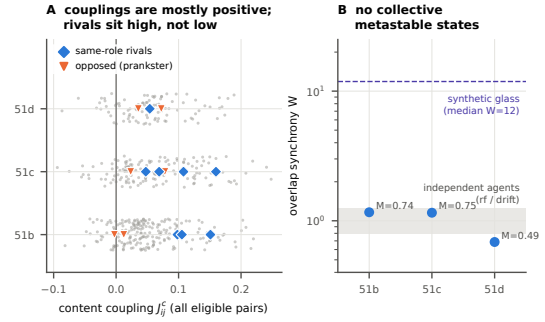
**Bottom line.** #51 is not a spin glass: private roles make it a random-field system with weak, mostly positive coupling, where rivals align rather than repel and the swarm never switches collectively.

level: hypothesis · pipeline checks 83% · updated 2026-10-04



### WHAT HAPPENED

Do #51's private, partly conflicting roles make the swarm a spin glass? Tested on sub-periods 51a-e (45 non-holdout days) against shared-objective weeks #38, #40, #44, after synthetic validation (the balance index separates factions from random signs; SK and a heterogeneous ferromagnet cannot be told apart under common drive). Content couplings are real in 51b and 51c ( $\rho_{\text{split}} = 0.46$ ,  $p = 0.003$ ) but 85–94% positive ( $\hat{\kappa} = 4\text{--}10$ ); the balance index is indeterminate. Same-role rivals co-move *more* than unrelated pairs ( $T_{\text{SR}} = +0.048$ ,  $p = 0.02$ ), the homophily rival. Positions persist ( $M = 0.5\text{--}0.75$ ) without collective switching ( $W \approx 1$ ). Contrast weeks are no less ferromagnetic; talk couplings sit at the noise floor. Holdout test (#51 tail) written, **not run**.



**Observables.** (a) Every eligible pair's content coupling in the counted #51 units (gray), with same-role rivals (blue) and Prankster vs Ethicist/Psychologist pairs (orange): mostly positive, rivals high. (b) Overlap synchrony  $W$  at the independent-agent level (gray band), far below a synthetic glass (dashed). High memory  $M$  without synchrony: agents hold their own positions; the swarm never jumps between collective states.

### PHYSICAL MODEL

Models 01 (SK) and 11 ( $O(n)$  spins). Spin  $\mathbf{s}_i$ : agent  $i$ 's whitened chat embedding per 30-min window ( $n = 32$ ). Fields: day field  $\mathbf{m}_d$ ; static field  $\mathbf{H}_i$  (role, lab, style).  $J_{ij}$ : correlation of within-day fluctuations  $\mathbf{x}_i, \mathbf{x}_j$  minus a cross-day surrogate;  $A, B, C$ : day folds.  $\kappa$ : SK ratio  $J_0/J$ ;  $\tau_3$ : gauge-invariant balance (SK  $\approx 0$ , factions  $\approx 0.5$ );  $q$ : overlap of day-demeaned states  $\hat{\delta}$ ;  $W$ : its synchrony beyond per-agent shifts.

$$P(\{\mathbf{s}\}) \propto \exp \left( \sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (\mathbf{m}_d + \mathbf{H}_i) \cdot \mathbf{s}_i \right)$$

$$J_{ij} = \text{corr}_w(\mathbf{x}_i, \mathbf{x}_j) - \text{corr}_w^{\text{surr}}(\mathbf{x}_i, \mathbf{x}_j)$$

$$\kappa = \sqrt{N} \bar{J} / \sigma_J, \quad \tau_3 = \text{tr}(J^A J^B J^C) / \text{tr}(J^A J^B)^{3/2}$$

$$q(d, d') = \frac{1}{N} \sum_i \hat{\delta}_{i,d} \cdot \hat{\delta}_{i,d'}$$

### SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Practically, little. “Spin glass” was the exciting reading of mixed-motive swarms: frustrated, history-dependent, hard to steer. The data say something duller and more usable. Assigned roles act as fixed fields that pin each agent's topic, the shared room adds a weak positive pull, and same-objective agents drift together. For an operator, roles, not inter-agent conflict, set what such a swarm talks about, and there are no hidden collective states to escape. Caveat: embeddings measure topic, not stance, so strategic opposition could be invisible here. The diagnostics port to other logs; the spin-glass vocabulary is not earned.

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RESULTS VS PREDICTIONS

Pending.

FAITHFULNESS SCORECARD

|                                                                                                                |   |                   |   |
|----------------------------------------------------------------------------------------------------------------|---|-------------------|---|
| A mapping                                                                                                      | 1 | F identifiability | 1 |
| B assumptions                                                                                                  | 1 | G ground truth    | 1 |
| C adequacy                                                                                                     | 1 | H comparative     | 0 |
| D unfitted predictions                                                                                         | 0 | I transfer        | 0 |
| E interventional                                                                                               | 0 |                   |   |
| Level: <b>hypothesis</b> . Scale 0 not done or failed, 1 partial, 2 passed ( <code>writeup/paper.tex</code> ). |   |                   |   |

CAVEATS

Pending.

GOAL PERIODS IN DETAIL

|      |     |              |
|------|-----|--------------|
| G38  | d   | descriptive  |
| G40  | d   | descriptive  |
| G44  | d   | descriptive  |
| G51a | d   | descriptive  |
| G51b | ×   | failed       |
| G51c | ~   | inconclusive |
| G51d | ~   | inconclusive |
| G51e | d   | descriptive  |
| G23  | ... | pending      |
| NE38 | ... | pending      |

ROUND 2

- **Essence.** Do conflicting incentives produce conflict behavior at all?
- **H22-R1** Cooperation default: agents help rivals at rates comparable to aligned pairs (Jev-labeled help acts).
- **H22-R2** A quantitative random-field fit (HH129): roles as fields plus weak positive coupling.

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